BEING SEPARATED IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author's closing phrase 'and we win' is quoted verbatim.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Base-change stability and Zariski locality of separatedness. Schemes, Lemmas schemes-lemma-separated-permanence and schemes-lemma-characterize-separated, invoked at proof lines 5804-5808. Supplies the two preliminary halves of fpqc locality.
- Universal closedness descends, and a universally closed immersion is a closed immersion. Same chapter, Lemma lemma-descending-property-universally-closed and Schemes, Lemma schemes-lemma-diagonal-immersion, invoked at proof lines 5815-5820. Gives that Delta is universally closed and, being an immersion, a closed immersion.
- The general principle for a property that is fpqc local on the base. Same chapter, Lemma lemma-descending-properties-morphisms, invoked at proof lines 5821-5822. Concludes the locality statement.

Source reference. Original lemma 02KP: sources/stacks-descent/pinned-original-source.tex, source lines 5624–5624. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KS: sources/stacks-descent/pinned-original-source.tex, source lines 5732–5732. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01KP: sources/stacks-chow/tags.txt, source lines 2058–2058. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01KJ: sources/stacks-chow/tags.txt, source lines 2052–2052. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01KU: sources/stacks-chow/tags.txt, source lines 2063–2063. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is separated" is fpqc local on the base.*

Proof. A base change of a separated morphism is separated, see Schemes, Lemma 01KU. Being separated is Zariski local on the base (from the definition or by Schemes, Lemma 01KP). Finally, let $S' \to S$ be a flat surjective morphism of affine schemes, and let $f : X \to S$ be a morphism. Assume that the base change $f' : X' \to S'$ is separated. This means that $\Delta' : X' \to X' \times_{S'} X'$ is a closed immersion, hence universally closed. Note that Δ' is the base change of $\Delta : X \to X \times_S X$ via $S' \to S$. By Lemma 02KS this implies Δ is universally closed. Since it is an immersion (Schemes, Lemma 01KJ) we conclude Δ is a closed immersion. Hence f is separated. Therefore Lemma 02KP applies and we win. □